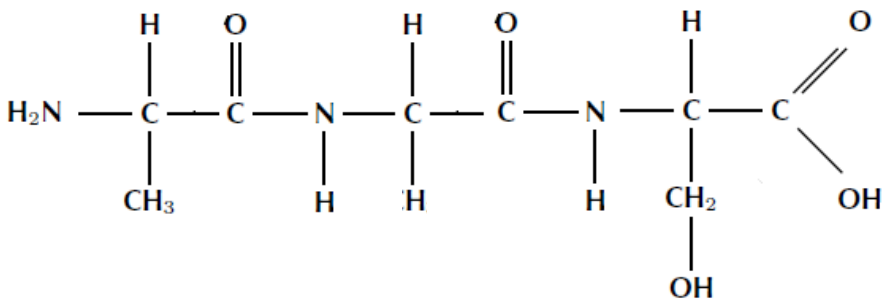
Section Multiple Choice Questions:
Proteins, Fats and Carbohydrate Past SAC (Chapters 15 and 16) Questions

Question 1

The structure of a molecule is represented below:



Which of the amino acids listed would not be produced when this molecule undergoes hydrolysis?

A. Alanine
B. Glycine
C. Threonine
D. Serine

Question 2

Which of the following fatty acids is the most saturated?

A. Linoleic
B. Palmitoleic
C. Oleic
D. Myristic

Question 3

Amino acids are classified as essential or non-essential. Which of the following statements is **true**?

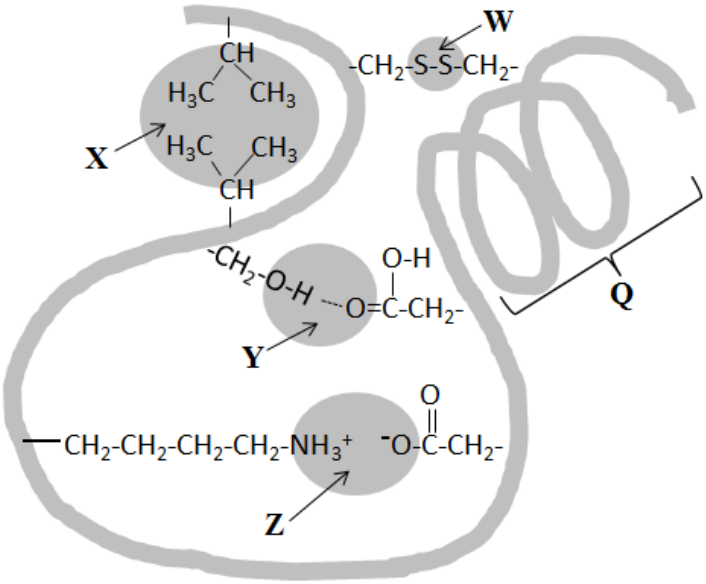
A. The Z-groups on essential amino acids are always polar.
B. Essential amino acids cannot be manufactured in the body and must be included in the diet.
C. Non-essential amino acids do not form bonds within the tertiary structure of a protein.
D. Non-essential amino acids are obtained through the foods we eat.

Question 4

Select the best definition for an enzyme.

A. An enzyme is an amino acid that speeds up chemical reactions.
B. An enzyme is a non-specific organic catalyst that speeds up chemical reactions.
C. Enzymes are selective proteins that speed up chemical reactions.
D. Enzymes are inorganic and speed up chemical reactions using the lock and key model.

Use the following information to answer Questions 5 - 7
The following diagram is a simplified representation of a section of a polymer.



Question 5
The types of bonding represented at X , Y and Z are:

	X	Y	Z
A.	dipole-dipole force	ionic bond	peptide link
B.	peptide link	dipole-dipole force	hydrogen bond
C.	dispersion forces	hydrogen bond	ionic bond
D.	dispersion forces	dispersion forcess	ionic bond

Question 6
The type of bonding represented at W, and the amino acids that were involved in the bonding are:

	bonding	amino acid 1	amino acid 2
A.	disulphide link	cysteine	cysteine
B.	glycosidic link	cysteine	cysteine
C.	ether link	cysteine	methionine
D.	peptide link	cysteine	methionine

Question 7
The section of polymer represented by Q forms in which structure, and is held together by which type of bonding?

	structure	type of bonding
A.	secondary	covalent bond
B.	secondary	hydrogen bond
C.	tertiary	covalent bond

D.	tertiary	hydrogen bond

Question 8

A fatty acid molecule contains 18 carbon atoms and 2 C=C double bonds. The fatty acid could be

A. $C_{18}H_{32}O_2$

B. Oleic acid

C. $C_{18}H_{36}O_2$

D. $C_{18}H_{32}COOH$

Question 9

A single strand of DNA contains 10 nucleotides. There are 24 hydrogen bonds between the strand and its complementary strand. The number of adenine **and** thymine bases in the strand must be.

A. 4

B. 5

C. 6

D. 7

Question 10

If an enzyme is produced from 1956 amino acid monomers, how many water molecules would be released?

A. 1956

B. 1014

C. 1955

D. 3912

Question 11

A nucleotide consists of

A. adenine, guanine, thymine, and cytosine

B. a purine base, a phosphate group, adenine and guanine

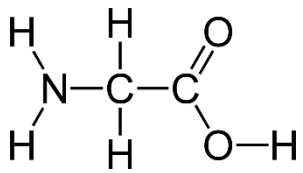
C. a deoxyribose sugar, a base, and a nitrate group

D. a phosphate group, a deoxyribose sugar group, and a purine base

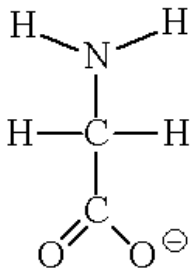
Question 12

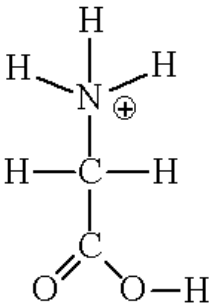
At pH 6, the amino acid glycine exists in its **zwitterion form**. Which of the following structures would represent glycine as it exists at pH 3?

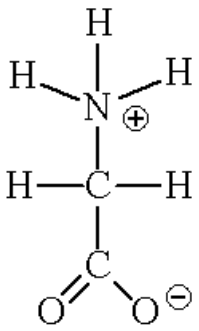
A.



C.



B. 

D. 

Question 13

The molecular formula of alanine is $C_3H_7O_2N$. If three alanine molecules react together to form a tripeptide, the relative molecular mass of this tripeptide would be:

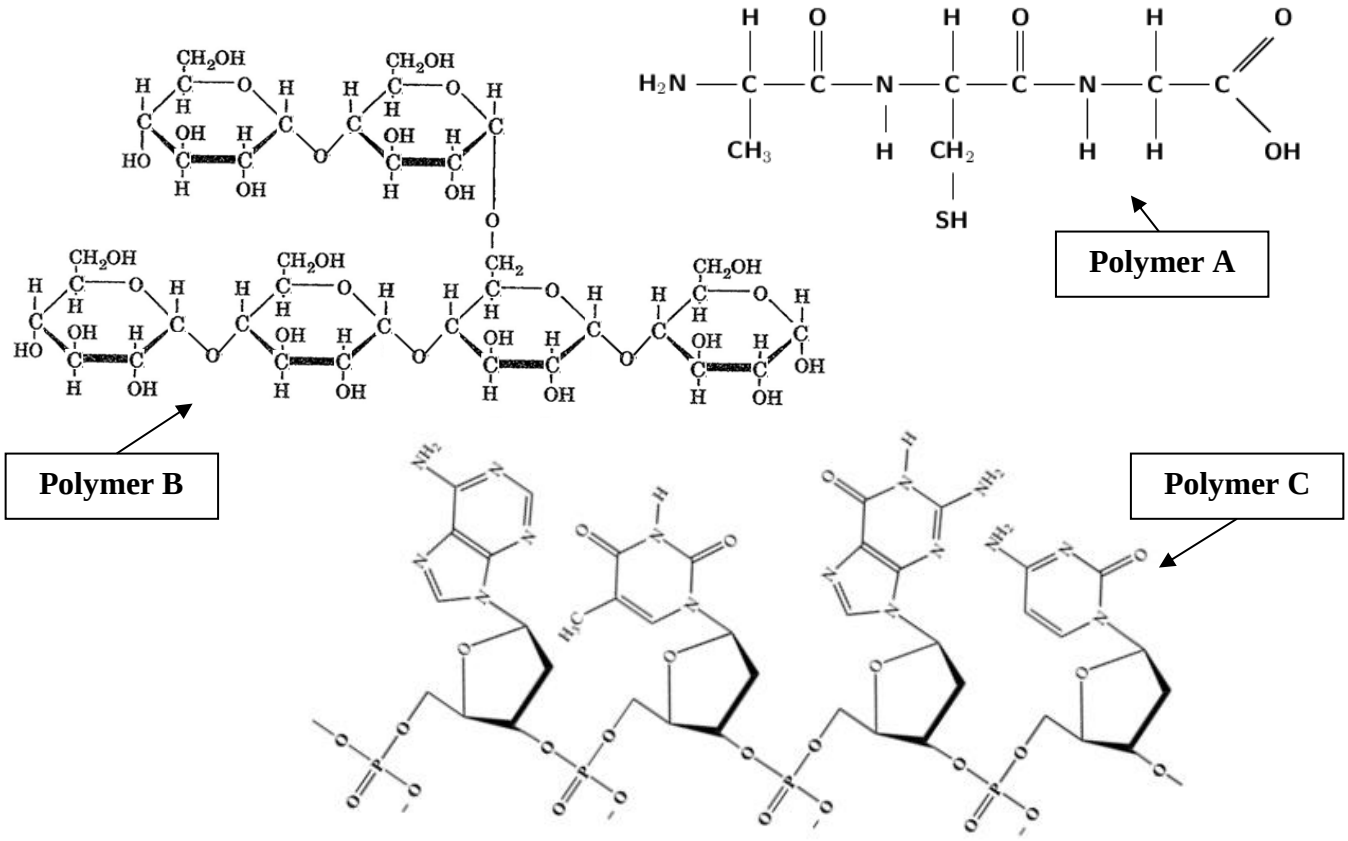
A. 89
B. 231
C. 267
D. 283

Section B Short Answer Questions:

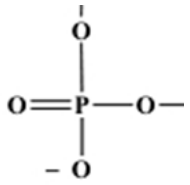
Proteins, Fats and Carbohydrate Past SAC (Chapters 15 and 16) Questions

Question 1

Many of the biomolecules found in our body are polymers. The structures of three such polymers are shown below:



a. Use polymers A, B, and C above to complete the following table:

	Polymer A	**Polymer B**	**Polymer C**
Type of biomolecule	Protein		
Name of link			Phosphodiester bond
Structure of link			
Number of monomers		6	
Structure and name of **one** monomer.	**Full** structure: Name:	**Full** structure: Name:	Use **symbols/words** to represent structure: Name:

11 marks

b. What type of reaction occurs between the monomers in order to produce polymer C?

_____

1 mark

c. Proteins and DNA both have secondary structures maintained by hydrogen bonding. The strength of bonding that results from the hydrogen bonds throughout the secondary structure of a protein is fairly consistent. This is not the case for DNA. Explain.

_____

_____

3 marks

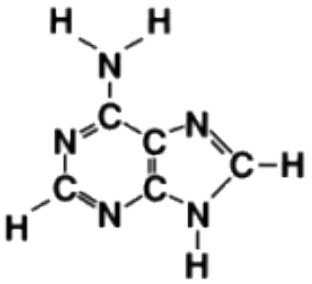
Total 15 marks

Question 2

Molecule X forms a component of DNA. It forms specific bonds with another nitrogen containing compound on a complimentary strand of DNA.

a. **Circle** the components on Molecule X that are directly involved in maintaining the **secondary structure** of DNA.

Molecule X

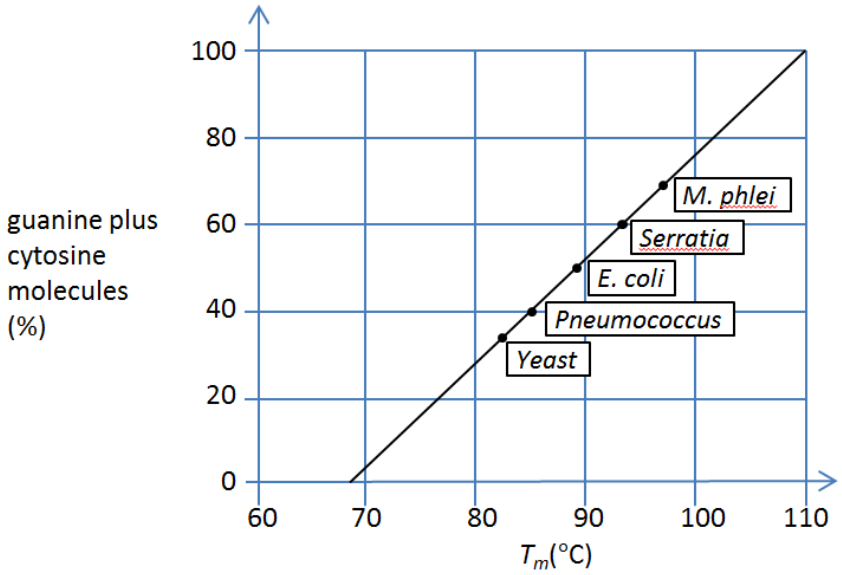


1 mark

b. Draw the structure of the base which is complimentary to Molecule X and circle the components of this molecule that are directly involved in maintaining the secondary structure of DNA with Molecule X.

2 marks

c. When double-stranded DNA is heated slowly, the double helix unravels to a random coil configuration over a narrow temperature range. The midpoint temperature of this process is known as the melting temperature (*Tm*). The graph below shows the *Tm* of various samples of DNA (taken from different living things) as a function of the base composition.



i. With reference to the structure of DNA, explain the trend depicted in the graph.

ii. In a sample of double stranded DNA from *E. coli* which contains 3000 base pairs, how many adenine and thymine molecules in total wold be present in a single strand of its DNA?

2 + 2 = 4 marks

d. In a cell, DNA strands are bonded tightly to proteins known as histones. These proteins contain a high proportion of the amino acids lysine and arginine. In the acidic environment of the DNA molecule, the side chain groups of these amino acids are charged.
Name the type of bonding that exists between histones and the DNA molecule, and explain how this bonding occurs between them.

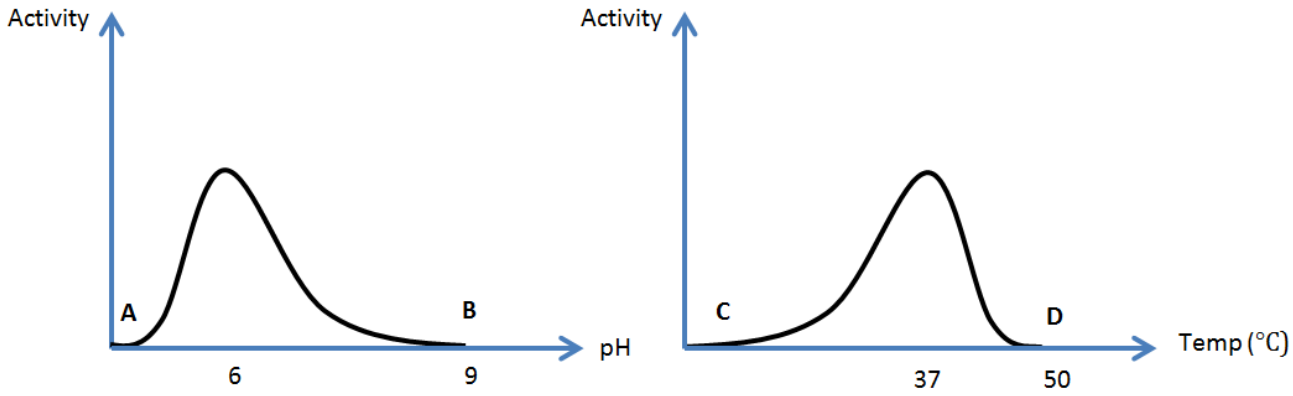
2 marks

Total 9 marks

Question 3

a. The graphs below illustrate the activity of an enzyme as a function of pH and temperature.

i. In which region of **each** of the graphs has protein denaturation occurred? Circle the appropriate answer(s) A, B, C, or D below.



ii. Explain why the catalytic function of an enzyme molecule is dependent on the way the molecule is folded.

_____

_____

_____

_____

_____

1 + 2 = 3 marks

b. Using full structural formulae, showing all bonds, write an equation to represent the reaction between alanine and proline to form the dipeptide ala-pro.

2 marks

Total 5 marks

Calorimetry Past SAC Questions

Question 4
The electrical calibration data for a solution calorimeter is shown below.

Volume of solution	**100.0 mL**
Initial temperature	**18.27°C**
Voltage	**6.10 V**
Current	**2.13 A**
Time	**10.0 minutes**
Final Temperature	**32.93°C**

a) Determine the calibration factor for this calorimeter

2 marks

b) When 0.370 g of zinc powder was added to 100.0 mL of 1.0 M aqueous copper (II) sulfate solution in the calorimeter, a 2.32°C temperature increase occurred.

(i) Write an ionic equation for the reaction between the zinc and the copper (II) solution.

1 mark

(ii) Calculate the change in enthalpy, ΔH, for this reaction.

4 marks
Total 7 marks